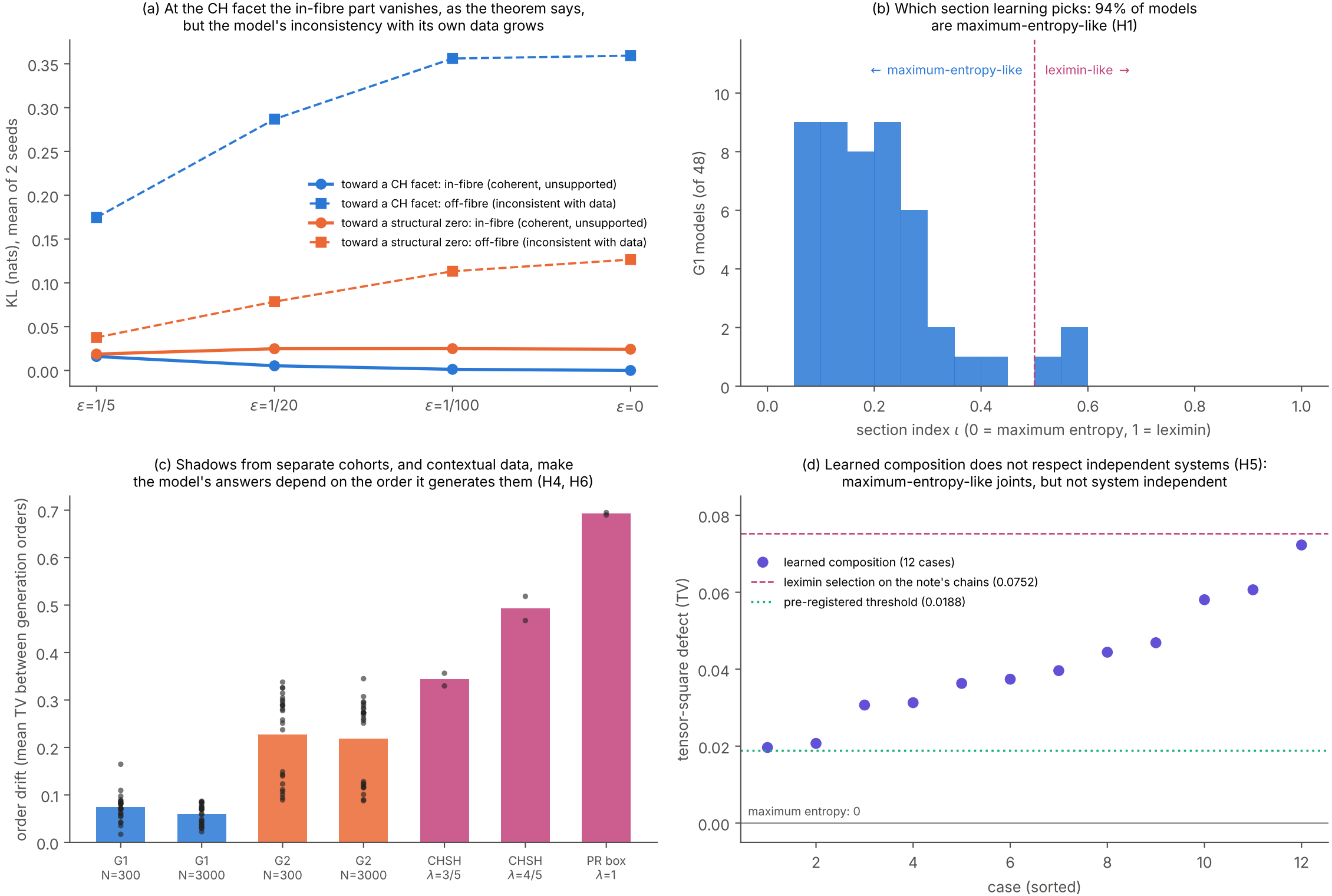
(a) At the CH facet the in-fibre part vanishes, as the theorem says, but the model's inconsistency with its own data grows
KL (nats), mean of 2 seeds
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
ε=1/5
ε=1/20
ε=1/100
ε=0
toward a CH facet: in-fibre (coherent, unsupported)
toward a CH facet: off-fibre (inconsistent with data)
toward a structural zero: in-fibre (coherent, unsupported)
toward a structural zero: off-fibre (inconsistent with data)
(b) Which section learning picks: 94% of models are maximum-entropy-like (H1)
← maximum-entropy-like
leximin-like →
G1 models (of 48)
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
section index ι (0 = maximum entropy, 1 = leximin)
(c) Shadows from separate cohorts, and contextual data, make the model's answers depend on the order it generates them (H4, H6)
order drift (mean TV between generation orders)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
G1 N=300
G1 N=3000
G2 N=300
G2 N=3000
CHSH λ=3/5
CHSH λ=4/5
PR box λ=1
(d) Learned composition does not respect independent systems (H5): maximum-entropy-like joints, but not system independent
tensor-square defect (TV)
0.00
0.02
0.04
0.06
0.08
learned composition (12 cases)
leximin selection on the note's chains (0.0752)
pre-registered threshold (0.0188)
maximum entropy: 0
2
4
6
8
10
12
case (sorted)